

Deepak Vungarala

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Ph.D. Candidate in Computer Engineering | SRC Research Scholar, ECE Department, New Jersey Institute of Technology

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Education

New Jersey Institute of Technology

Ph.D. in Computer Engineering, CGPA: 4.0/4.0

Jan. 2024 – Present

Expected December 2027

New Jersey Institute of Technology

M.S. in Electrical and Electronics Engineering, CGPA: 3.5/4.0

Oct. 2022 – Dec. 2023

Research and Professional Experience

New Jersey Institute of Technology

Research Assistant, ACAD Lab — Advisor: Dr. Shaahin Angizi

Jan. 2024 – Present

Newark, NJ

- Conduct research on LLM- and agent-based design automation for AI accelerators, analog in-memory computing, and silicon photonic integrated circuits.
- Author/co-author of 15+ publications in venues including IEEE TCAD, ISCAS, MLCAD, ICLAD, and GLSVLSI.
- Develop and evaluate automated hardware-design workflows spanning architecture generation, circuit optimization, simulation, and implementation.

Cognichip Inc.

Research Intern

Sept. 2025 – Aug. 2026

Hybrid

- Developed LLM-based and hardware-aware problem-formulation methods for chip-design tasks, including graph-based techniques and data curation, and translated research prototypes into design-automation workflows for industry applications.

New Jersey Institute of Technology

Teaching Assistant / Instructor

Spring 2024 – Summer 2025

Newark, NJ

- Instructed ECE-349 Digital Systems Lab and ECE-429 Computer Communications Lab; designed laboratory exercises in digital design, FPGA programming, and communication protocols, and mentored student hardware projects.

New Jersey Institute of Technology

Research Assistant, VLSI Lab — Advisor: Dr. Durga Misra

Summer 2023

Newark, NJ

- Characterized RRAM devices for neuromorphic computing using Keysight analyzers and probe stations, with emphasis on resistive-switching behavior for emerging accelerator memories.

Publications

- [1] S. Fall, D. Vungarala, and S. Angizi, “G-DiffPS: Physics-Informed Graph Diffusion Policy for Amortized Multi-Topology RF Phase Shifter Synthesis,” *ACM/IEEE MLCAD*, 2026. Accepted.
- [2] D. Vungarala, D. Najafi, A. Aljoudi, Z. Ghanaatian, N. Khoshavi, G. Datta, A. Roohi, M. Nikdast, and S. Angizi, “PICasso: An AI-Enabled Design Framework for Autonomous Optimization of Silicon Photonic Devices,” *IEEE ICLAD*, 2026. Accepted.
- [3] D. Vungarala, S. Fall, A. Roohi, R. Zand, and S. Angizi, “RL-AmpSyn: Reinforcement Learning-Enhanced Operational Amplifier Generation and Optimization with LLMs,” *IEEE FCCM*, 2026. Accepted.
- [4] D. Vungarala, K. Pandit, G. Aragonda, J. McLynch, A. Adeoye-Davids, B. Galecio, N. H. Phan, A. Khreishah, R. Zand, A. Ghosh, and S. Angizi, “Closing the Loop in LLM-Based Hardware Generation: An Autonomous Agentic Workflow for Robust TPU Design,” *IEEE VTS*, 2026. Accepted.
- [5] M. Nazzal, K. Nguyen, D. Vungarala, R. Zand, S. Angizi, H. Phan, and A. Khreishah, “FedChip: Federated LLM for Artificial Intelligence Accelerator Chip Design,” *IEEE ICLAD*, 2025. arXiv:2508.13162.
- [6] D. Vungarala, M. H. Amin, P. Mercati, A. Ghosh, A. Roohi, R. Zand, D. Z. Pan, and S. Angizi, “LIMCA: LLM for Automating Analog In-Memory Computing Architecture Design Exploration,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2026. arXiv:2503.13301.
- [7] D. Vungarala, M. E. Elbittity, K. Pandit, S. Syed, S. Alam, A. Ghosh, R. Zand, and S. Angizi, “TPU-Gen: LLM-Driven Custom Tensor Processing Unit Generator,” *IEEE ICLAD*, 2025. arXiv:2503.05951.
- [8] D. Vungarala, M. H. Amin, P. Mercati, A. Roohi, R. Zand, and S. Angizi, “LLM-IMC: Automating Analog In-Memory

Computing Architecture Generation with LLMs,” *IEEE FCCM*, 2025.

- [9] D. Vungarala, S. Alam, A. Ghosh, and S. Angizi, “SPICEPilot: AI-Guided SPICE Code Generation,” *IEEE ICRC*, 2024. arXiv:2410.20553.
- [10] D. Vungarala, M. Nazzal, M. Morsali, C. Zhang, A. Ghosh, A. Khreishah, and S. Angizi, “SA-DS: A Dataset for LLM-Driven AI Accelerator Design Generation,” *IEEE ISCAS*, 2025. arXiv:2404.10875.
- [11] D. Vungarala, M. H. Amin, A. Roohi, A. Ghosh, R. Zand, and S. Angizi, “From Prompt to Accelerator: A Perspective on LLM-Based Analog In-Memory Accelerator Design Automation,” *ACM GLSVLSI*, 2025.
- [12] G. Aragona, D. Najafi, D. Vungarala, S. Tabrizchi, A. Roohi, and S. Angizi, “Maximizing Sub-Array Resource Utilization in Digital Processing-in-Memory,” *ACM GLSVLSI*, 2025.
- [13] D. Vungarala, M. Morsali, S. Tabrizchi, A. Roohi, and S. Angizi, “Comparative Study of Low Bit-width DNN Accelerators: Opportunities and Challenges,” *IEEE MWSCAS*, 2023, pp. 797–800.
- [14] D. Vungarala, “Gen-acceleration: AI-based Hardware Accelerator Design,” Master’s Thesis, New Jersey Institute of Technology, 2023.
- [15] C. Wang, D. Vungarala, K. Navarro, N. Adwani, and T. Han, “Pinch Sensor: An Input Device for In-Hand Manipulation with the Index Finger and Thumb,” *IEEE/ASME AIM*, 2023.

Academic Service and Mentoring

- **Peer Review:** IEEE/ACM ICCAD, DATE, DAC, ASP-DAC, ICLAD, and ISCAS.
- **Undergraduate Mentoring:** Mentored ACAD Lab undergraduate and REU researchers on hardware-design and AI-for-EDA projects.

Technical Skills

Programming	Python, C, VHDL, Chisel HDL, Scala, SPICE/HSPICE
Tools	OpenROAD, Synopsys Design Compiler, ModelSim, Mentor Graphics, MATLAB, PySpice, Linux
Hardware	Keysight Semiconductor Analyzer, Cascade MicroTech probe station, Oscilloscope
Research Areas	Large Language Models for EDA, In-Memory Computing, Silicon Photonic IC Design, ASIC Design
Languages	English, Telugu, Hindi, Spanish (Beginner)

Awards, Certifications, and Activities

- **Awards:** NSF Student Award (MWSCAS 2023); Best Technical Presentation; Smart India Hackathon Finalist (2021); two-time State Boxing Gold Medalist.
- **Certifications:** Programming with Python and Python Data Structures (University of Michigan, Coursera); Problem Solving in C (NPTEL); IoT Workshop (IIT Hyderabad).
- **Leadership and Service:** Former National Kickboxing Team Captain; five national Kung-fu medals; certified referee for national competitions; organized NJIT Engineers’ Day with 650+ participants.